POINTWISE CHARACTERIZATIONS OF GEOMETRIC NORMALITY BY PURELY INSEPARABLE BASE CHANGE AND BY THE LOCAL RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and closes the cycle of equivalences; no step is referred to the reader.

Editorial note. The counted spans carry 5 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-geometrically-normal-at-point is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Geometric normality of a k-algebra. Algebra, Definition algebra-definition-geometrically-normal and Lemma algebra-lemma-geometrically-normal, invoked at proof lines 2128-2130 and 2144-2146. Supplies the algebraic characterization used in the middle equivalence.
- Spectrum of a purely inseparable base change and localness of the tensor product. Algebra, Lemmas algebra-lemma-p-ring-map and algebra-lemma-characterize-local-ring, invoked at proof lines 2138-2141. Supplies that the tensor product with a finite purely inseparable extension is local with the same spectrum.
- Surjectivity is stable under base change. Morphisms, Lemma morphisms-lemma-base-change-surjective, invoked at proof lines 2149-2152. Supplies a point above x for an arbitrary field extension.

Source reference. Original definition 0380: sources/stacks-chow/tags.txt, source lines 4219–4219. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00E9: sources/stacks-chow/tags.txt, source lines 526–526. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 037Z: sources/stacks-chow/tags.txt, source lines 4218–4218. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 0BRA: sources/stacks-chow/tags.txt, source lines 14763–14763. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01S1: sources/stacks-chow/tags.txt, source lines 2322–2322. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. Let $x \in X$. The following are equivalent*

(1) *X is geometrically normal at x,*
(2) *for every finite purely inseparable field extension k' of k and $x' \in X_{k'}$ lying over x the local ring $\mathcal{O}_{X_{k'},x'}$ is normal, and*
(3) *the ring $\mathcal{O}_{X,x}$ is geometrically normal over k (see Algebra, Definition 0380).*

Proof. It is clear that (1) implies (2). Assume (2). Let k'/k be a finite purely inseparable field extension (for example $k = k'$). Consider the ring $\mathcal{O}_{X,x} \otimes_k k'$. By Algebra, Lemma 0BRA its spectrum is the same as the spectrum of $\mathcal{O}_{X,x}$. Hence it is a local ring also (Algebra, Lemma 00E9). Therefore there is a unique point $x' \in X_{k'}$ lying over x and $\mathcal{O}_{X_{k'},x'} \cong \mathcal{O}_{X,x} \otimes_k k'$. By assumption this is a normal ring. Hence we deduce (3) by Algebra, Lemma 037Z.

Assume (3). Let k'/k be a field extension. Since $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is surjective, also $X_{k'} \to X$ is surjective (Morphisms, Lemma 01S1). Let $x' \in X_{k'}$ be any point lying over x. The local ring $\mathcal{O}_{X_{k'},x'}$ is a localization of the ring $\mathcal{O}_{X,x} \otimes_k k'$. Hence it is normal by assumption and (1) is proved. □